

Renu Jakkampudi

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PROFESSIONAL SUMMARY

As a Data Science Master's student I focus on turning complex datasets into actionable business and research insights. From building high-accuracy computer vision models at NUS to architecting automated data pipelines for AI campaigns at Gabriel AI, I specialize in the end-to-end data lifecycle. I use Python, SQL, and advanced modeling to deliver validated, high-impact solutions that prioritize data integrity and measurable results.

EDUCATION

M.S. in Data Science | Arizona State University, Tempe, AZ
B.S. in Computer Science | Vellore Institute of Technology, India

GPA: 4.0/4.0 | May 2026
GPA: 3.72/4.0 | May 2024

EXPERIENCE

Gabriel AI | Data Engineer Intern
Tempe, Arizona

September 2025 - January 2026

- Designed and automated backend data flows for phone-based AI campaigns, handling number routing, validation, and API integrations that improved workflow reliability.
- Developed and maintained services that use Gemini to format and generate voicemail content with consistent structure and tone, helping the team create campaigns faster and with fewer manual steps.

National University of Singapore | ML & Data Science Intern
Singapore

December 2022 - November 2023

- Developed a medical alert system for real-time patient emotion detection using CNN, BiLSTM, and MobileNet, achieving up to 99.94% accuracy on a 28K+ image dataset.
- Researched Conditional GANs for anime-style image generation by converting COCO 2017 images using AnimeGAN, analyzing training stability and performance limits.

Indian Space Research Organization (ISRO) - ADRIN | Machine Learning Engineer Intern
Hyderabad, India

May 2023 - June 2023

- Preprocessed Massachusetts Roads aerial dataset by converting TIFF files to JPEG and generating low-resolution Gaussian-blurred variants, creating paired datasets for supervised super-resolution.
- Trained SRCNN, LAPSRN, and AdaptSRCNN models, achieving 34.05 dB PSNR and restoring high-resolution details for geospatial mapping.

PROJECTS

Trusted AI - Evidence First Health Q&A with Graph Constrained RAG

November 2025

- Improved retrieval quality by +15% Recall@10 and +12% nDCG@10 using BM25, dense retrieval, cross-encoder reranking, and UMLS/RxNorm graph checks.
- Added a safety layer (emergency triage + prescription refusal), achieving 95% citation completeness, <5% hallucination rate, and >90% safety-interception accuracy.

Spotify Music Trend Analysis

May 2025

- Analyzed 32K+ tracks to identify key drivers of popularity (valence, energy, danceability) using PCA, KMeans clustering, and Random Forest.
- Built an interactive Tableau dashboard to visualize genre trends, explicit content patterns, release behavior, and top-artist insights.

CineCurate - Hybrid RL based Movie Recommender Model

May 2024

- Designed a hybrid recommender integrating epsilon-greedy RL, collaborative filtering, and content-based filtering for personalized movie suggestions.
- Tuned RL exploration - exploitation strategies using offline experiments to improve recommendation relevance and system stability.

TECHNICAL SKILLS

Languages: Python, SQL, R, Bash.

ML & Deep Learning: Scikit-learn, TensorFlow, PyTorch, Keras, HuggingFace Transformers, Pandas, NumPy, PySpark.

ML Domains: Deep Learning, NLP, Computer Vision, LLMs, Agentic Workflow, RAG Pipelines, Prompt Engineering, Recommendation Systems, Reinforcement Learning, Vector Embeddings, AI Evaluation, Model Benchmarking.

MLOps & Infrastructure: MLflow, Docker, Kubernetes, CI/CD for ML, AWS (S3, Lambda), Azure, Distributed Training

Databases: Snowflake, PostgreSQL, MySQL, Unit testing.

Certifications: Google Advanced Data Analytics Professional Certificate | Oracle Machine Learning Autonomous DB Specialist.